

Find the general solution of each differential equation.

$$1) \frac{dy}{dx} = \frac{x^3}{y^2}$$

$$2) \frac{dy}{dx} = \frac{1}{\sec^2 y}$$

$$3) \frac{dy}{dx} = 3e^{x-y}$$

$$4) \frac{dy}{dx} = \frac{2x}{e^{2y}}$$

$$5) \frac{dy}{dx} = \frac{2x}{y^2}, y(2) = \sqrt[3]{13}$$

$$6) \frac{dy}{dx} = 2e^{x-y}, y(-3) = \ln \frac{3e^3 + 2}{e^3}$$

$$7) \frac{dy}{dx} = \frac{1}{\sec^2 y}, y(3) = 0$$

$$8) \frac{dy}{dx} = \frac{e^x}{y^2}, y(-1) = \frac{\sqrt[3]{e^3 + 3e^2}}{e}$$

$$9) \frac{dy}{dx} = -\frac{1}{\sin y}, y(3) = \frac{\pi}{2}$$

$$10) \frac{dy}{dx} = \frac{2x}{e^{2y}}, y(2) = \frac{\ln 5}{2}$$

$$11) \frac{dy}{dx} = \frac{3x^2}{e^{2y}}, y(1) = 0$$

$$12) \frac{dy}{dx} = \frac{1+x^2}{y^2}, y(-1) = -\sqrt[3]{4}$$

$$1 + C_1 y = \sqrt[3]{\dots}$$

$$2) \tan y = x + C$$

$$y = \tan^{-1}(x + C)$$

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$$y = \ln(3e^x + C)$$

$$4) \frac{e^{2y}}{2} = x^2 + C_1$$

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$$5) y = x - 3 \quad y = \cos^{-1}(x - 3), 2 < \dots$$

$$11) \frac{e^{2y}}{2} = x^3 - \frac{1}{2}, x > \frac{\sqrt{6}}{2}$$

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$$5) y = x - 3 \quad y = \cos^{-1}(x - 3), 2 < \dots$$

$$11) \frac{e^{2y}}{2} = x^3 - \frac{1}{2}, x > \frac{\sqrt{6}}{2}$$

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Find the general solution of each differential equation.

$$1) \frac{dy}{dx} = \frac{x^3}{y^2}$$

$$2) \frac{dy}{dx} = \frac{1}{\sec^2 y}$$

$$3) \frac{dy}{dx} = 3e^{x-y}$$

$$4) \frac{dy}{dx} = \frac{2x}{e^{2y}}$$

$$5) \frac{dy}{dx} = \frac{2x}{y^2}, y(2) = \sqrt[3]{13}$$

$$6) \frac{dy}{dx} = 2e^{x-y}, y(-3) = \ln \frac{3e^3 + 2}{e^3}$$

$$7) \frac{dy}{dx} = \frac{1}{\sec^2 y}, y(3) = 0$$

$$8) \frac{dy}{dx} = \frac{e^x}{y^2}, y(-1) = \frac{\sqrt[3]{e^3 + 3e^2}}{e}$$

$$9) \frac{dy}{dx} = -\frac{1}{\sin y}, y(3) = \frac{\pi}{2}$$

$$10) \frac{dy}{dx} = \frac{2x}{e^{2y}}, y(2) = \frac{\ln 5}{2}$$

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$$12) \frac{dy}{dx} = \frac{1+x^2}{y^2}, y(-1) = -\sqrt[3]{4}$$

$$1 + C_1 y = \sqrt[3]{\dots}$$

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